

Cor Pulmonale Revisited

From Ferrer and Harvey to the Present

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Abstract

The term *cor pulmonale* has traditionally been used as a synonym for right heart failure due to chronic respiratory diseases, although this condition is less frequently seen in the modern era because of the use of long-term oxygen therapy along with aggressive measures directed at optimizing ventilation and gas exchange. The mechanisms by which adaptation or maladaptation of right heart structure and function in the broader setting of pulmonary vascular disease, either intrinsic to the pulmonary circulation or due to

respiratory diseases, have garnered considerable interest along with the development of medical and surgical treatments for pulmonary hypertension. Thus, the right heart is no longer considered an “innocent bystander” in pulmonary hypertension, but rather a key component in its pathophysiology. Furthermore, the status of right heart function is a major determinant of outcome. Accordingly, the right heart has become a potential, appealing target for novel therapies to treat hypertensive pulmonary vascular disease.

Keywords: *cor pulmonale*; right heart; pulmonary hypertension

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The term *cor pulmonale*, coined by Paul Dudley White nearly a century ago, had long been used as a surrogate term for right ventricular failure (1). Although its use has fallen by the wayside, the development of sophisticated invasive and noninvasive techniques has enabled us to explore right heart structure and function and the pulmonary circulation in a detailed, sophisticated manner. This article reviews the origins of the term *cor pulmonale*, the seminal work performed in the 1950s and 1960s by Cournand and associates that first explored the structure and function of the right ventricle in health and disease, and more recent work that has provided a greater understanding of the unique properties of the right heart.

In his state-of-the-art review, Fishman defined *cor pulmonale* as “a synonym for pulmonary heart disease...used to signify right ventricular enlargement from disorders that affect either the structure or function of the lungs” (2). He defined the

elements of *cor pulmonale* as follows: 1) the etiology of the heart disease may be either intrinsic pulmonary disease, including abnormalities in the pulmonary vessels, or inadequate function of the chest bellows or inadequate ventilatory drive from the respiratory centers; 2) the cardiomegaly is confined predominantly, if not exclusively, to the right ventricle and may take the form of dilatation, hypertrophy, or both; 3) pulmonary arterial hypertension is the *sine qua non*, and whether dilatation or hypertrophy contributes more importantly to the right ventricular hypertrophy depends on the degree and duration of the pulmonary hypertension; and 4) neither congenital heart disease nor acquired disease of the left side of the heart can be implicated as initiating mechanisms for the pulmonary arterial hypertension. Using the current World Health Organization classification of pulmonary hypertension, conditions associated with *cor pulmonale* would

include some, but not all, of those conditions under the headings of group 1 (pulmonary artery hypertension), group 3 (pulmonary hypertension resulting from chronic respiratory disease and/or hypoxia), group 4 (chronic thromboembolic pulmonary hypertension), and at least partially group 5 (miscellaneous causes). Excluded would be group 2, in which pulmonary hypertension is the result of left-sided heart disease, and congenital heart disease (listed under group 1).

Working in the cardiopulmonary research unit at Bellevue Hospital in New York under the direction of future Nobel Prize in Medicine laureates André Cournand and Dickinson Richards, Irené Ferrer and Réjane Harvey first characterized several clinical phenotypes of *cor pulmonale*, particularly focusing on acute and chronic *cor pulmonale* resulting from chronic lung disease (3, 4). They showed that the clinical features of acute *cor pulmonale*, typically the result of acute

exacerbations of chronic lung disease, may be reversible with time and general treatment measures for the underlying cause. They also noted that, although left-sided heart failure is associated with reduced cardiac output, cor pulmonale is characterized by normal or even increased cardiac output, owing to the increased myocardial fiber stretch resulting from increased intravascular volume. Only late in the course of chronic cor pulmonale does overt right heart failure with decreased cardiac output ensue. Nearly 60 years later, a large number of studies have emphasized the importance of right ventricular stiffness, increased afterload, ventriculoarterial uncoupling, and right ventricular dyssynchrony as the key features of right heart failure due to advanced pulmonary artery hypertension (5–11). The latter mechanism is also thought to contribute to altered left ventricular systolic and diastolic dysfunction due to ventricular interdependence that occurs in the setting of right ventricular pressure overload (12).

At the cellular level, fatty deposits have been found in the right ventricular myocardium of patients with right ventricular pressure overload (13), although their meaningfulness is unknown, because even a “failing” right ventricle can return to normal function if afterload is normalized, such as after lung transplantation for pulmonary hypertension or pulmonary endarterectomy for chronic thromboembolic

pulmonary hypertension. Studies using uptake of radiolabeled fluorodeoxyglucose as an index of efficiency of myocardial metabolism have shown impaired uptake in the right ventricular myocardium of patients with pulmonary hypertension (13). Interestingly, when fluorodeoxyglucose uptake improves or even normalizes in patients with pulmonary artery hypertension treated with epoprostenol, they achieve hemodynamic improvement; in contrast, untreated and epoprostenol-treated patients who showed no improvement in fluorodeoxyglucose uptake failed to improve hemodynamics as well. Thus, therapies that target the imbalance between energy substrate supply and demand or efficiency of substrate utilization may be novel approaches to the treatment of conditions associated with right ventricular pressure overload.

Although therapeutic options were limited during their lifetime, Ferrer and Harvey also investigated the effects in cor pulmonale of therapies that had been used to treat conditions affecting the left heart, such as digoxin and isoproterenol. With the advent of the pulmonary artery hypertension–targeted therapeutic era, considerable interest has developed in studying the effects of these therapies on right heart structure and function, using physiologic and imaging studies. For example, several studies employing echocardiography have shown that targeted medical therapy for pulmonary artery

hypertension is associated with reduction in right ventricular size and eccentricity index and improvement in the right ventricular–left ventricular end-diastolic ratios, indicative of increased pulmonary blood flow and improved left ventricular filling (14, 15, 16). Magnetic resonance imaging has been used to assess the effects of therapy on both right heart structure and function, including assessments of right ventricular mass, stroke volume, and ejection fraction (17, 18). Because right ventricular function is the main determinant of outcome in pulmonary artery hypertension, measurements that reliably assess changes in right heart function may be adopted as clinically relevant endpoints in future clinical trials of novel therapies for pulmonary hypertension in the future.

The pioneering work by Ferrer and Harvey brought attention to the importance and unique characteristics of right ventricular function in pulmonary vascular disease. While the right ventricle has hitherto been an unexplored target for treatment of pulmonary vascular disease, we now have the opportunity to focus on the key pathophysiological mechanisms of right heart dysfunction as part of a comprehensive approach to the treatment of disorders of the pulmonary circulation. ■

Author disclosures are available with the text of this article at www.atsjournals.org.

References

- 1 Dankmeijer J, Herles F, Ibrahim M, Reid DD, Richards DW, Stuart-Harris CH, *et al.*; World Health Organization. Chronic cor pulmonale: report of an expert committee. WHO Technical Report Series No. 213. *Circulation* 1963;27:594–615.
- 2 Fishman AP. State of the art: chronic cor pulmonale. *Am Rev Respir Dis* 1976;114:775–794.
- 3 Ferrer MI, Harvey RM, Cathcart RT, Webster CA, Dickinson WR, Cournand A. Some effects of digoxin upon the heart and circulation in man: digoxin in chronic cor pulmonale. *Circulation* 1950;1:161–186, illust.
- 4 Harvey RM, Ferrer MI, Cournand A. The treatment of chronic cor pulmonale. *Circulation* 1953;7:932–940.
- 5 Hagger D, Condliffe R, Woodhouse N, Elliot CA, Armstrong IJ, Davies C, *et al.* Ventricular mass index correlates with pulmonary artery pressure and predicts survival in suspected systemic sclerosis–associated pulmonary arterial hypertension. *Rheumatology (Oxford)* 2009;48:1137–1142.
- 6 van Wolferen SA, Marcus JT, Boonstra A, Marques KM, Bronzwaer JG, Spreeuwenberg MD, *et al.* Prognostic value of right ventricular mass, volume, and function in idiopathic pulmonary arterial hypertension. *Eur Heart J* 2007;28:1250–1257.
- 7 Rain S, Handoko ML, Trip P, Gan CT, Westerhof N, Stienen GJ, *et al.* Right ventricular diastolic impairment in patients with pulmonary arterial hypertension. *Circulation* 2013;128:2016–2025, 1–10.
- 8 Guilhaire J, Haddad F, Boulate D, Decante B, Denault AY, Wu J, *et al.* Non-invasive indices of right ventricular function are markers of ventricular–arterial coupling rather than ventricular contractility: insights from a porcine model of chronic pressure overload. *Eur Heart J Cardiovasc Imaging* 2013;14:1140–1149.
- 9 Vonk-Noordegraaf A, Haddad F, Chin KM, Forfia PR, Kawut SM, Lumens J, *et al.* Right heart adaptation to pulmonary arterial hypertension: physiology and pathobiology. *J Am Coll Cardiol* 2013;62(25, Suppl):D22–D33.
- 10 Vonk-Noordegraaf A, Westerhof BE, Westerhof N. The relationship between the right ventricle and its load in pulmonary hypertension. *J Am Coll Cardiol* 2017;69:236–243.
- 11 van de Veerdonk MC, Kind T, Marcus JT, Mauritz G-J, Heymans MW, Bogaard HJ, *et al.* Progressive right ventricular dysfunction in patients with pulmonary arterial hypertension responding to therapy. *J Am Coll Cardiol* 2011;58:2511–2519.
- 12 Gan C, Lankhaar JW, Marcus JT, Westerhof N, Marques KM, Bronzwaer JG, *et al.* Impaired left ventricular filling due to right-to-left ventricular interaction in patients with pulmonary arterial hypertension. *Am J Physiol Heart Circ Physiol* 2006;290:H1528–H1533.
- 13 Brittain EL, Talati M, Fessel JP, Zhu H, Penner N, Calcutt MW, *et al.* Fatty acid metabolic defects and right ventricular lipotoxicity in human pulmonary arterial hypertension. *Circulation* 2016;133:1936–1944.
- 14 Oikawa M, Kagaya Y, Otani H, Sakuma M, Demachi J, Suzuki J, *et al.* Increased [18 F]fluorodeoxyglucose accumulation in right ventricular

- free wall in patients with pulmonary hypertension and the effect of epoprostenol. *J Am Coll Cardiol* 2005;45:1849–1855.
- 15 Kuehne T, Yilmaz S, Steendijk P, Moore P, Groenink M, Saaed M, *et al.* Magnetic resonance imaging analysis of right ventricular pressure–volume loops: *in vivo* validation and clinical application in patients with pulmonary hypertension. *Circulation* 2004;110:2010–2016.
 - 16 Galiè N, Hinderliter AL, Torbicki A, Fourme T, Simonneau G, Pulido T, *et al.* Effects of the oral endothelin-receptor antagonist bosentan on echocardiographic and Doppler measures in patients with pulmonary arterial hypertension. *J Am Coll Cardiol* 2003;41:1380–1386.
 - 17 Peacock AJ, Crawley S, McLure L, Blyth K, Vizza CD, Poscia R, *et al.* Changes in right ventricular function measured by cardiac magnetic resonance imaging in patients receiving pulmonary arterial hypertension–targeted therapy: the EURO-MR study. *Circ Cardiovasc Imaging* 2014;7:107–114.
 - 18 Hinderliter AL, Willis PW IV, Barst RJ, Rich S, Rubin LJ, Badesch DB, *et al.*; Primary Pulmonary Hypertension Study Group. Effects of long-term infusion of prostacyclin (epoprostenol) on echocardiographic measures of right ventricular structure and function in primary pulmonary hypertension. *Circulation* 1997;95:1479–1486.